

Explainable flood damage assessment using multi-atrous self-attention and vision-language integration

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ABSTRACT

Flood disasters triggered by excessive rainfall cause severe damage to infrastructure and pose significant risks to human life. Within the context of disaster management, accurately identifying affected structures and providing interpretable analytical results are of critical importance. This study proposes a new disaster analysis framework that integrates the Multi-Atrous Self-Attention (MASA) mechanism, which is designed to capture multi-scale spatial features effectively, with vision-language models for explainable flood assessment. The proposed approach consists of two main components. The first component performs segmentation to detect and quantify flood-affected structures, while the second component employs a fine-tuned vision language model to generate natural language descriptions of the disaster scene. The MASA module processes image-mask pairs from the FloodNet dataset to segment disaster related structures, whereas the LoRA (Low Rank Adaptation) enhanced BLIP-2 (Bootstrapped Language Image Pre-training) model learns image-text pairs from the LADI-v2 dataset to produce textual disaster descriptions. Through this dual stage structure, the system provides both quantitative and linguistic outputs, enabling interpretable flood impact assessment. Experimental results demonstrate that the proposed MASA-based segmentation model achieves a mean Intersection over Union (mIoU) of 73.78 % on FloodNet, outperforming state-of-the-art segmentation models. Furthermore, the LoRA-fine-tuned BLIP-2 model achieves a BLEU score of 80.77 % on the LADI-v2 dataset, indicating fluent, contextually relevant, and semantically coherent textual outputs. The proposed system contributes to disaster analysis by enhancing explainability and interpretability in flood damage assessment.

1. Introduction

Natural disasters are devastating events that pose serious threats to both human life and property security. These events lead to significant economic consequences that affect communities across the world. Rapid and accurate access to disaster related information is of critical importance for ensuring the effectiveness of emergency response operations and minimizing overall losses. The immediate identification of disaster affected regions directly influences the success of search and rescue operations as well as the timely delivery of humanitarian aid.

In recent years, innovative technologies developed to meet this need have made substantial contributions to disaster response efforts (Wieland et al., 2025). Among natural disasters, floods are one of the most destructive types, often triggered by sudden and intense rainfall. They can cause severe loss of life and large scale material damage across vast areas. In densely populated and urbanized regions, flash floods can

lead to infrastructure collapse, disruption of transportation networks, and the evacuation of affected zones (Karimov and Sheshukov, 2017). The ability of search and rescue teams to allocate resources effectively and initiate lifesaving operations in a timely manner largely depends on this capability (Hasnaoui et al., 2025). Artificial intelligence (AI) based approaches in disaster management have been utilized to detect various disaster types such as floods, earthquakes, and wildfires (Ijaz et al., 2023; Kyrkou and Theodorides, 2020; Mustafa et al., 2024; Shianios et al., 2024). In this context, high resolution images obtained from Unmanned Aerial Vehicles (UAVs) serve as a valuable data source for the automatic identification of disaster types and the spatial analysis of their impacts. Image based analysis also plays a critical role in assessing the spatial distribution, severity, and affected elements of disasters such as wildfires (Feng et al., 2025; Shianios et al., 2025), post-tsunami, hurricane damage (Li et al., 2023a), earthquakes (Ekkazan and Karsligil, 2025; Karakas et al., 2025; Liu et al., 2022; Wiguna et al., 2024), and

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floods (Guclu et al., 2025; Huang et al., 2024; Li et al., 2023a).

Floods, in particular, highlight the need for rapid information generation and accurate decision making in disaster management. Recent studies have demonstrated the effectiveness of deep learning based image analysis for post flood damage assessment. Some studies have investigated building damage assessment in post flood scenarios using attention enhanced convolutional neural networks and DeepLabV3-based semantic segmentation models to classify and segment structural damage from RGB imagery (Wu et al., 2024a, 2024b). Attention-enhanced convolutional neural networks have been employed to improve the discrimination of damage patterns by emphasizing structurally relevant features, while DeepLabV3-based semantic segmentation models have been developed to achieve fine-grained extraction of cracks, spalling, and other structural defects from RGB imagery. These approaches demonstrate the effectiveness of pixel level and region level analysis for post-disaster building damage evaluation. Rahne-moonfar et al. (2021) proposed a deep learning based segmentation approach using high resolution aerial imagery to assess structural damage and predict flood extent in urban settings. Shi et al. (2024) introduced an interactive prompt-based semantic segmentation method for flood mapping. Their approach improves segmentation accuracy by integrating spatial attention modules with prompt encoders that incorporate points, curves, boxes, and masks. However, the study focuses solely on segmentation and does not infer information about the affected areas or disaster magnitude. The image-based flood depth estimation approaches have been proposed to infer water level severity in urban environments by exploiting visual cues and attention mechanisms within CNN architectures (Wu et al., 2025). However, this method does not provide pixel-level flood segmentation or interpretable, multimodal descriptions that support comprehensive disaster situation awareness. Shaheen et al. (2025) combined the strengths of Swin Transformer and SegFormer architectures to detect flood regions in UAV imagery with high accuracy. Although the study achieves better results than standard transformer-based models through multi scale feature extraction, its computational complexity remains high. Román et al. (2025) developed a real-time 3D mapping and damage assessment framework for a flash flood in Spain by integrating UAV-based LiDAR, multispectral imagery, and gas sensor data. The study utilized YOLOv11 for object detection and support vector machines to analyze structural damage and environmental effects.

Sundaresan and Solomon (2024) proposed a DeepLabV3+ based segmentation model using high-resolution UAV imagery. Experiments conducted on the FloodNet dataset demonstrated that the ResNet-50 backbone achieved the highest accuracy and mIoU. The proposed model integrates Atrous Spatial Pyramid Pooling (ASPP) and atrous convolutions to capture multi-scale spatial features, enabling pixel-level flood detection. Islam et al. (2023) proposed a deep learning based classification model for post flood region identification and aid prioritization using UAV images. Experiments using InceptionV3 and DenseNet achieved 83 % accuracy in classifying flood severity. Additionally, the Merge Sort algorithm was applied to optimize aid distribution by prioritizing distance and impact level. La Salandra et al. (2025) developed a UAV based sensing system named RivAir to determine river surface area and flow velocity in real time. The study employed YOLOv8 and optical flow algorithms for onsite data processing and achieved over 90 % accuracy in water surface detection. This confirms the value of AI-driven UAV systems as an effective tool for flood and water management. Munawar et al. (2021) presented a comprehensive review of image processing and machine learning approaches for flood management and introduced an integrated classification framework bridging the two domains. Their analysis showed that combining remote sensing, SAR, and UAV imagery with deep learning substantially improves flood detection and mapping; however, studies specifically targeting post disaster analysis remain limited. Simantiris and Panagiotakis (2024) introduced an unsupervised flood segmentation approach for UAV imagery that derives flood masks using LAB color cues, vegetation indices,

and edge-based features. While the method avoids the need for labeled training data, its color driven nature limits its reliability in complex urban environments, particularly under shadow and reflection conditions. Pech-May et al. (2024) proposed a U-Net-based approach for flood detection using Sentinel-1 synthetic aperture radar (SAR) satellite imagery. For this purpose, the Sen1Floods11 dataset was employed, and flood-affected areas were identified from satellite images. However, the dataset used in the study is binary, containing only background and flood classes. The ability of U-Net to capture relevant features in SAR imagery was evaluated in the study. Şener and Ergen (2025) proposed a new encoder-decoder-based segmentation approach to accurately delineate flood inundation areas. To reduce information loss, a specialized intermediate module and skip connections were incorporated into the model. However, the study focuses solely on flood detection from UAV imagery and does not generate any textual reports from the segmentation outputs. In addition, the proposed architecture has a higher number of parameters compared to existing segmentation models.

Situ et al. (2025) proposed an attention-based hybrid deep learning framework for urban flood risk assessment. In their study, flood depth was estimated using an LSTM-SegNet model enhanced with multi-head self-attention (MSA), followed by detailed land-use classification using a DeepLab-based model integrated with EfficientNet-Lite4 and squeeze-and-excitation (SE) modules. In the final stage, expected annual damage (EAD) was calculated at a spatial scale using depth-damage curves. However, the study does not focus on explainability aspects such as visual interpretability, textual description generation, or natural language interpretation of flood damage. Şener et al. (2024) proposed a CNN-based segmentation model that combines hybrid attention mechanisms and atrous convolution modules to identify flood affected regions. Although these approaches achieve high accuracy in binary (flood/background) segmentation, they provide only visual outputs and do not offer any explainability components or language-based analysis. Table 1 presents a comparative summary of recent key studies on flood detection and segmentation.

As observed in Table 1, although significant progress has been made in the field of flood detection and segmentation, most studies primarily focus on segmentation and do not generate semantic or linguistic explanations for disaster scenes. Moreover, in many studies, only the flooded areas are segmented, without providing information about objects affected by the flood, such as buildings, roads, bridges, and trees.

In this study, a new approach is proposed to support post disaster emergency response and to determine the extent of damage in residential areas based on acquired imagery. To this end, a framework that combines advanced segmentation with explainable artificial intelligence is presented for flood detection using aerial and satellite images. The core segmentation network employs a MASA mechanism to effectively capture multi-scale spatial features and global context, thereby achieving highly accurate segmentation. While ASPP modules facilitate the precise extraction of fine details, the self-attention structure enhances contextual perception between semantically related yet spatially distant regions.

For vision linguistic explanation generation, a BLIP-2 + LoRA-based approach is adopted to provide an innovative solution to the limitations identified in the literature. In this study, BLIP-2 and LoRA were selected due to their advantages in parameter efficiency and adaptability compared to other vision language models. BLIP-2 connects a frozen CLIP-based visual encoder and a frozen large language model through a lightweight Q-Former, enabling cross-modal alignment without the need for full end-to-end fine-tuning. This design significantly reduces computational cost while preserving generalization capability, particularly in disaster scenarios where domain shifts are common. Additionally, LoRA provides an efficient fine-tuning mechanism by introducing low-rank updates to selected projection matrices, allowing the model to adapt to disaster specific datasets with only a small number of trainable parameters.

Table 1

Literature studies on flood detection and segmentation.

References	Dataset	Methodology	Key Results	Limitations
Shaheen et al. (2025)	FloodNet (UAV)	Combined Swin Transformer + SegFormer Architecture	Multi-Scale Feature Extraction	Lack of Vision-Based Multimodal Analysis and Interpretation
Román et al. (2025)	UAV + LiDAR	YOLOv11 + SVM + Sensor Fusion	3D mapping, environmental pollutant, and structural damage analysis were performed.	Limited real-time processing; high computational cost; lack of linguistic explanation.
Sundaresan and Solomon (2024)	FloodNet (UAV)	DeepLabV3+ (ResNet50, ASPP)	Successful segmentation was achieved with 92 % accuracy and 71 % mIoU.	Segmentation only; lack of explainability and multimodal analysis.
Islam et al. (2023)	Custom Flood Image Dataset	CNN (Inception V3, DenseNet) + Merge Sort	Flood severity classification and aid prioritization were performed with 83 % accuracy.	Limited to visual classification; lacks segmentation and explainability.
La Salandra et al. (2025)	UAV River Images	YOLOv8 + Farneback Optical Flow	Real-Time Water Surface Segmentation	Limited to flow detection; lacks linguistic or semantic analysis.
Munawar et al. (2021)	Various (Satellite + UAV Data)	Image Processing + Machine Learning Review (ANN, SVM, ANFIS, etc.)	The necessity of hybrid models in disaster management has been emphasized.	Post-disaster analyses are limited; no deep learning integration.
Simantiris and Panagiotakis (2024)	Unlabeled UAV Images	Unsupervised Color-Based Segmentation	Flood detection was achieved without requiring labeled data.	No deep feature extraction; only color and texture information used; limited interpretability.

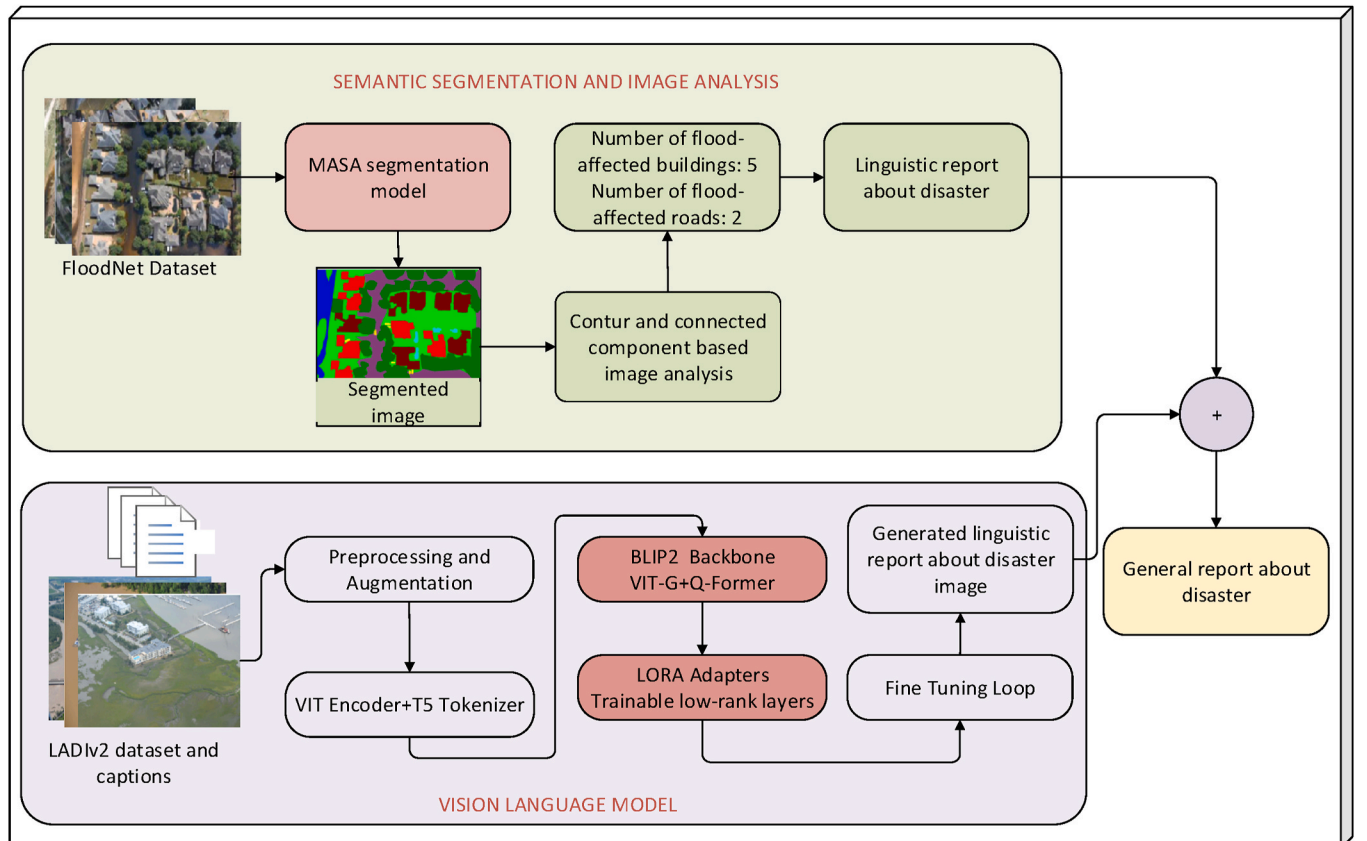
The proposed method performs high-accuracy flood segmentation on the FloodNet dataset, while producing natural-language descriptions of the images using the LADI-v2 dataset, thus offering both quantitative and interpretable information for disaster management. By leveraging the synergy between deep learning and vision language models, this framework introduces a multimodal, explainable, and decision-support-oriented perspective to disaster analysis.

The main contributions of this study are summarized as follows.

- Enhanced feature representation through the integration of ASPP and self-attention.

- Robust segmentation performance in complex, detail-sensitive flood scenarios.
- Natural language-based interpretability that maps visual outputs to user understandable descriptions.

Overall, the proposed approach represents an important step toward more effective artificial intelligence for disaster management by simultaneously improving accuracy and explainability in flood mapping systems.

**Fig. 1.** Architecture of the proposed MASA-BLIP2 LoRA-based disaster analysis framework.

2. Materials and methods

In this study, a two-stage deep learning approach is proposed for detecting flood-affected regions and generating explainable textual information about the disaster scene. In the first stage, high-resolution UAV images from the FloodNet dataset are used to perform segmentation. Through the segmented outputs, quantitative information about flood-affected structures such as buildings, roads, and other infrastructure elements is obtained.

In the second stage, the LADI-v2 dataset is utilized to train the BLIP-2 + LoRA model using image-text pairs. This model is fine-tuned to generate natural language based disaster descriptions directly from visual inputs.

Finally, the numerical results obtained from the segmentation stage are integrated with the textual explanations produced by the BLIP-2 + LoRA model, generating a human-readable analytical report for each disaster scene. The overall architecture of the proposed system is illustrated in Fig. 1.

As illustrated in Fig. 1, the proposed framework employs two different datasets for complementary tasks. The FloodNet dataset is used exclusively for semantic segmentation, where the MASA model produces pixel-level flood maps and numerical descriptors such as the number of flooded buildings and roads, extracted through contour and component based analyses. In parallel, the LADI-v2 dataset comprising image caption pairs is used to train the BLIP-2 LoRA vision language model, which independently processes the original disaster image and generates a linguistic description. The segmentation outputs are not used as

direct inputs to BLIP-2; instead, the numerical summaries derived from MASA are combined with the textual output of BLIP-2 to produce a unified and comprehensive disaster report. This configuration ensures that each module is optimized for its specific modality while contributing complementary information to the final analysis.

2.1. The proposed MASA model for semantic segmentation

A new semantic segmentation model extending the DeepLabV3+ framework (Chen et al., 2018) is proposed to detect flood-affected structures from imagery. To enhance segmentation performance in visually complex flood scenes, a Non-Local Self-Attention mechanism is integrated into the model. This structure captures long range dependencies beyond conventional convolutional limits, enabling clearer recognition of small and irregular objects. The model architecture employs a ResNet-50 backbone for deep feature extraction and processes input images with a resolution of 512×512 pixels. To capture multi-scale contextual information, an ASPP module with varying dilation rates is incorporated, allowing simultaneous analysis of both large-scale flood regions and local fine details. The self-attention component complements ASPP by modeling global contextual relationships and improving the understanding of object interactions within the scene. To evaluate the effectiveness of different dilation rates used in the ASPP module, ablation experiments were conducted. The results demonstrate that the combination of ASPP and self-attention significantly enhances segmentation accuracy, particularly in structurally complex flood imagery. The overall architecture of the MASA-based

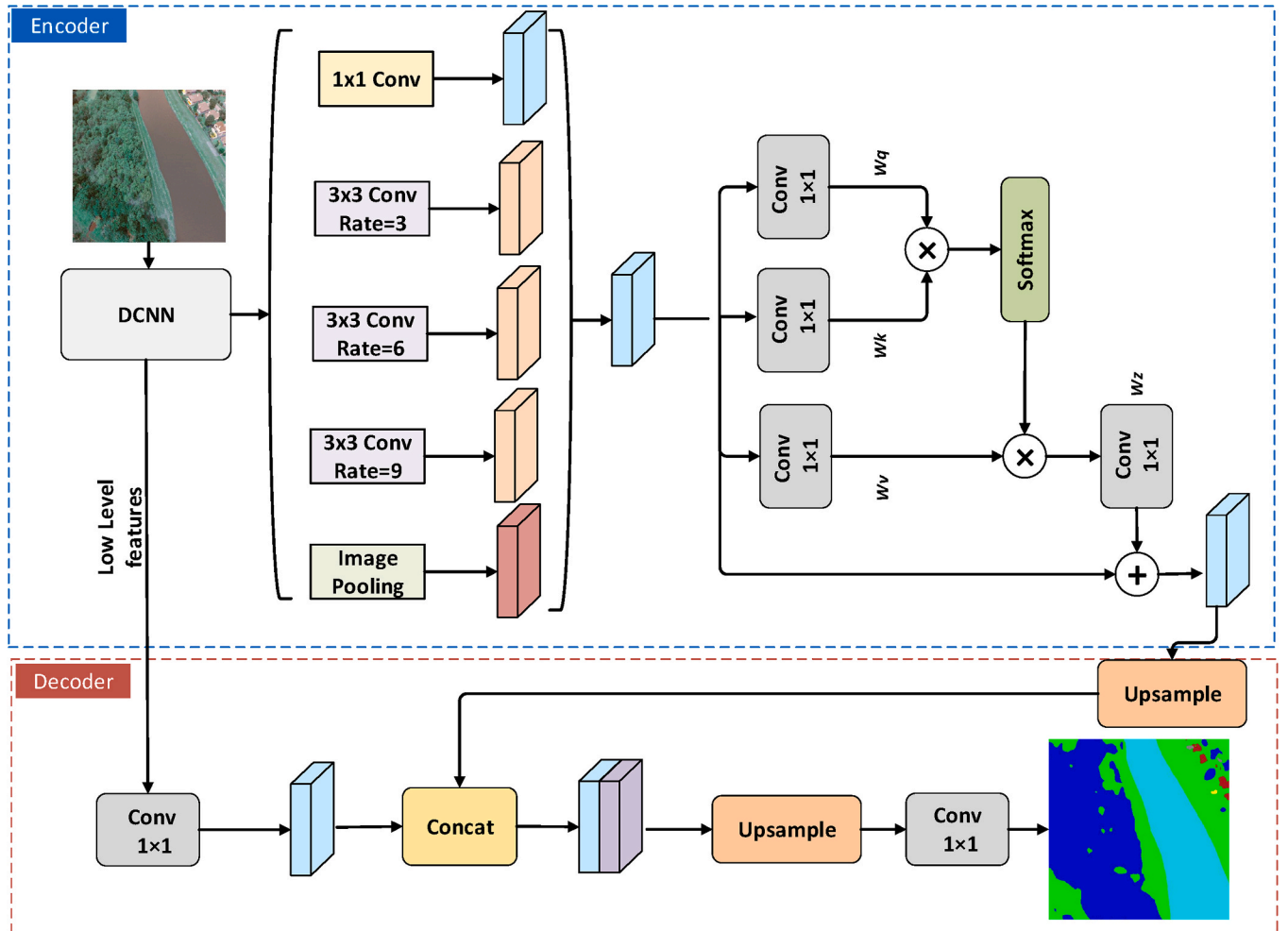


Fig. 2. Architecture of the proposed MASA model.

segmentation model is illustrated in Fig. 2.

The architecture in Fig. 2 consists of two main components: an encoder and a decoder. To extract deep features from the input images, the model employs a ResNet-50 backbone, which has been pre-trained on the ImageNet dataset. Owing to its deep residual learning structure, ResNet-50 is highly effective in capturing high-level semantic representations from visual data (He et al., 2015).

The ResNet-50 architecture includes residual connections that help mitigate the vanishing gradient problem commonly encountered in deep neural networks. This design not only preserves trainability as the network depth increases but also enables more efficient representation learning in deeper layers (Guclu and Akin, 2025). When F denotes a convolutional layer, the residual connections in the ResNet-50 architecture can be mathematically expressed as shown in Equation (1).

$$y = F(x, \{W_i\}) + x \quad (1)$$

In Equation (1), x represents the input tensor, while y denotes the output tensor. $F(x, \{W_i\})$ refers to the block function composed of convolutional layers and their associated parameters. These residual connections enhance the network's learning capacity by reducing information loss in deeper layers. While intermediate features in the model provide high-level semantic representations, the lower-level features contribute to the preservation of spatial details. In the first stage of the model, the feature map Z is obtained as shown in Equation (2).

$$Z = \text{ResNet50}(X), Z \in \mathbb{R}^{32 \times 32 \times 1024} \quad (2)$$

In Equation (2), X denotes the input image, $\text{ResNet50}()$ represents the feature extraction function, and Z corresponds to the resulting feature map with spatial dimensions 32×32 and 1024 channels. To optimize the trade-off between segmentation accuracy and computational efficiency, the proposed MASA architecture employs a truncated ResNet-50 backbone. Specifically, the encoder utilizes the feature extraction layers up to layer 3, omitting the final layer 4 block. This strategic modification serves two primary purposes: First, it maintains a higher spatial resolution (1/16 of the input size instead of 1/32) at the output of the backbone, which is crucial for preserving the fine-grained boundary details of flood-affected structures. Second, it significantly reduces the model's complexity by discarding the parameter-heavy final residual block. Consequently, the total number of trainable parameters is reduced to approximately 17.86 million. The loss of high-level semantic context typically provided by layer 4 is effectively compensated by the subsequent Multi-Atrous Module (MAM) and Self-Attention blocks, which capture global dependencies with a lower computational footprint. After capturing low-level features, this layer constructs the ASPP structure using 3×3 atrous convolution blocks with different dilation rates, combined with image pooling operations. This structure enables multi-scale feature extraction by considering both local and global

formulation, where query and key projections reduce the channel dimension to one-eighth of the input using 1×1 convolutions, while the value projection preserves the original channel size. This implicit channel adjustment reduces computational complexity while maintaining rich semantic information. The attention response is added to the input feature map through a residual connection, enhancing global contextual awareness before decoder upsampling and feature fusion with low-level encoder features.

The output of the self-attention module is merged with low-level features from the encoder and transferred to the decoder. In the decoder stage, 1×1 convolution, upsampling, and concatenation operations are applied to recover fine spatial details, ultimately enabling accurate segmentation of flood-affected regions.

To effectively capture multi-scale contextual information, the proposed segmentation model integrates an ASPP module adapted from the DeepLabV3+ architecture (Chen et al., 2017). This module combines parallel convolutional layers with different dilation rates and global average pooling, enabling the network to extract contextual information across objects of varying scales. Through dilated convolutions, the ASPP module enhances robustness to object scale variations and improves segmentation accuracy (Yu and Koltun, 2015). The standard convolution operation is mathematically formulated in Equation (3).

$$y[i, j] = \sum_{m=-k}^k \sum_{n=-k}^k x[i + m, j + n] \cdot w[m, n] \quad (3)$$

In Equation (3), $y[i, j]$ represents the output value, $x[i, j]$ denotes the input value, and $w[m, n]$ refers to the convolution kernel weight (Guclu et al., 2024). The parameter k denotes the half-size of the convolution kernel. Conversely, the dilated convolution operation is expressed using a dilation rate r and is mathematically formulated as shown in Equation (4).

$$y[i, j] = \sum_{m=-k}^k \sum_{n=-k}^k x[i + r + m, j + r + n] \cdot w[m, n] \quad (4)$$

In Equation (4), the dilation rate (r) represents the distance between the elements that make up the convolution kernel. As this rate increases, the receptive field of the convolution operation expands, allowing information to be gathered from a larger spatial area without a significant increase in computational cost. Consequently, the model can capture broader contextual features of the objects within the image.

The feature map (Z) obtained in Equation (2) is then fed into the ASPP module, which performs convolution operations with different dilation rates to capture multi-scale contextual information. As described in Equation (5), this process enables the simultaneous evaluation of image details at multiple scales, leading to the generation of more robust and semantically rich feature maps.

$$Z_{ASPP} = \text{Concat}[\text{Conv}_{1 \times 1}(Z), \text{Conv}_{3 \times 3}^{(r=3)}(Z), \text{Conv}_{3 \times 3}^{(r=6)}(Z), \text{Conv}_{3 \times 3}^{(r=9)}(Z), U(\text{AvgPool}(Z))] \quad (5)$$

contexts. The features obtained from the ASPP module are then passed to the self-attention module, which models dependencies between spatially distant but semantically related regions, thereby enhancing the network's global contextual awareness.

In the proposed MASA architecture, the self-attention module is integrated after the MAM at the encoder-decoder transition stage, operating on high-level feature maps extracted by the ResNet encoder. Specifically, the attention block receives the MAM output with 256 channels, where semantic abstraction is maximized and spatial resolution is reduced, making it suitable for long-range dependency modeling. The self-attention mechanism follows a single-head non-local attention

In Equation (5), U represents the upsampling operation performed using the bilinear interpolation method. The innovative aspect of the proposed model lies in the Self-Attention block integrated into the conventional convolutional neural network structure. This block establishes relationships between spatially distant yet semantically related regions, enhancing the model's ability to capture long-range dependencies.

Inspired by non-local network architectures, this mechanism models long-range interactions to improve global contextual awareness (Wang et al., 2018). The attention map is generated using the query-key-value (Q, K, V) approach and combined with the input features to strengthen

both spatial and semantic consistency in segmentation.

This mechanism computes, on a global scale, the relationship between a feature at any given point in the image and all other points. The feature map is first flattened into the form $Z_{flat} \in \mathbb{R}^{(1024 \times 256)}$. Subsequently, the query (Q), key (K), and value (V) matrices are computed through learnable weights, as formulated in Equation (6).

$$Q = Z_{flat}w_q, K = Z_{flat}w_k, V = Z_{flat}w_v \quad (6)$$

Here, the matrices w_q , w_k , and w_v represent the parameters learned by the network. Finally, the attention output is computed as shown in Equation (7). The resulting feature map is then restored to its original spatial dimensions and combined with the output from the ASPP module.

$$Z_{out} = Z_{ASPP} + \text{Reshape}(Z_{attn}), Z_{out} \in \mathbb{R}^{32 \times 32 \times 256} \quad (7)$$

In (7), Z_{ASPP} represents the feature map produced by the spatial pyramid pooling module, Z_{attn} denotes the attention-enhanced feature representation, and Z_{out} is the final fused feature map obtained by adding Z_{ASPP} and the reshaped attention output. The resulting feature map is fused with lower level, detail rich outputs from the ResNet-50 layers. These combined features are further refined through a sequence of convolutional layers. In the final stage, the feature map is upsampled to the input image resolution using bilinear interpolation to produce a class-based segmentation map. The high-resolution segmentation maps generated by the proposed approach are intended for use in structural damage detection and automated disaster-reporting workflows.

To accurately segment buildings, roads, and water bodies in complex flood scenarios, the model integrates convolutional layers with varying dilation rates and attention mechanisms to capture multi-scale contextual information. For continuity preservation and noise removal, a morphological opening operation with a five-pixel elliptical structuring element is applied to maintain the integrity of damaged regions. Connected-component analysis is then performed to label clusters, and

clusters with areas below a predefined threshold are discarded. Fig. 3 illustrates the sequential image-processing steps applied to the segmentation model's predicted mask.

Fig. 3 first assigns a class identifier to each pixel, after which the class-specific mask for flood-affected buildings is constructed. Contour extraction and a morphological opening operation are applied to reduce noise and sharpen object boundaries. Using connected-component analysis, contiguous pixel clusters are detected and the geometric centroid of each component is computed (He et al., 2017). The numerical values derived from the mask are then converted into a textual report.

Connected-component analysis operates on a binary image according to the principle of grouping neighboring pixels that belong to the same class. Each connected component represents a set of pixels corresponding to a particular object or region. A connected component is defined as shown in Equation (8).

$$C_i = \{(x, y) | B(x, y) = 1, \exists (x', y') \in N(x, y) \rightarrow (x', y') \in C_i\} \quad (8)$$

In Equation (8), $B(x, y)$ denotes the binary segmentation mask, $N(x, y)$ represents the neighborhood of pixel (x, y) . The parameter C_i is the connected component that contains all pixels (x, y) with $B(x, y) = 1$ such that at least one neighboring pixel (x', y') also belongs to the same component. The area of each component A_i is computed according to Equation (9). The centroid of each component is then calculated using moment-based formulas as given in Equation (10).

$$A_i = \sum_{(x, y) \in C_i} 1 \quad (9)$$

$$(cx_i, cy_i) = (\sum_{(x, y) \in C_i} x / A_i, \sum_{(x, y) \in C_i} y / A_i) \quad (10)$$

In (10), (cx_i, cy_i) denotes the centroid of the connected component C_i . The parameters x and y are the pixels coordinates within C_i . The centroids are used to characterize the spatial distribution and statistical properties of structures affected by flooding. Connected-component

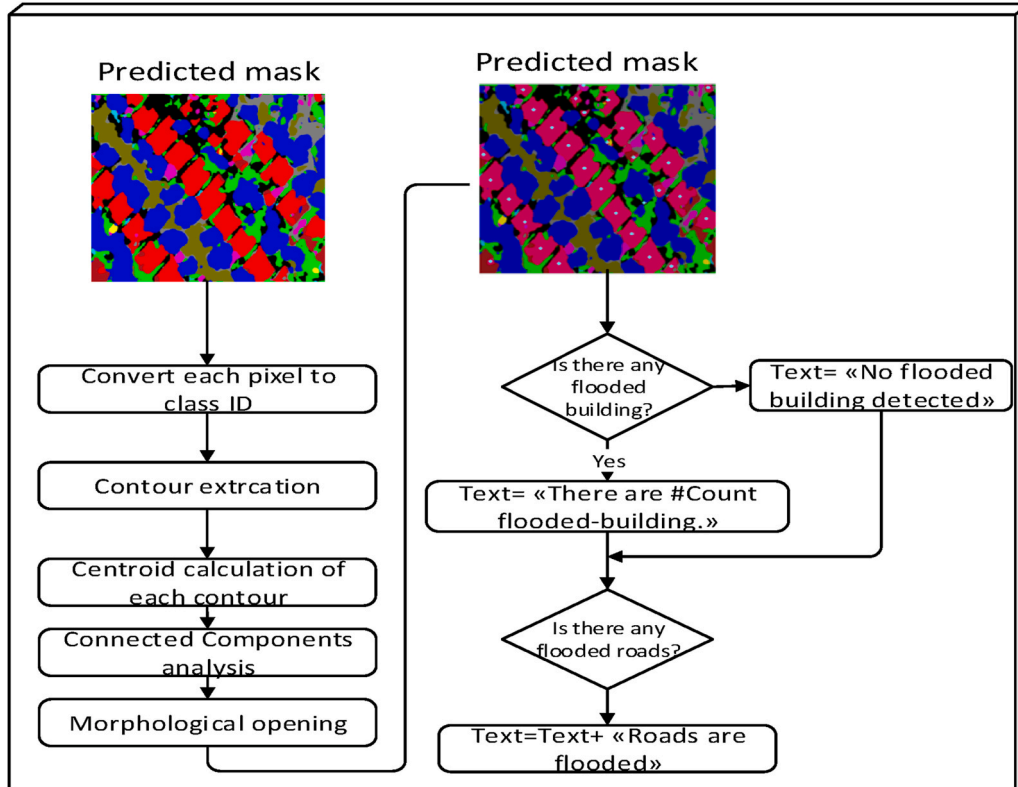


Fig. 3. Image-processing and connected-component analysis pipeline for detecting flood-affected buildings.

analysis enables independent enumeration of each object in the post-segmentation masks and the localization of their centroids.

2.2. Vision-to-linguistic information generation using the LoRA-Enhanced BLIP-2 model

To generate descriptive texts related to flood disasters, a model optimized with a LoRA-enhanced BLIP-2 mechanism is proposed. The proposed model enables lightweight fine-tuning by reducing the number of trainable parameters required for retraining multimodal models through the integration of LoRA. The architecture consists of three main components: a visual encoder, a query transformer with LoRA adapters, and a language decoder.

In addition to pixel-level classification performed on the FloodNet dataset, this study also aims to provide semantic and textual explanations of disaster scenes. While FloodNet is suitable for segmentation of classes such as buildings, roads, and water bodies, the LADI-v2 dataset is employed in this stage to provide paired image-text data describing disaster scenarios. The overall architecture of the proposed system is illustrated in Fig. 4.

In Fig. 4, for text generation from UAV images containing flood scenes, the LoRA-enhanced BLIP-2 model first processes the data through the AutoProcessor component, where images are converted into pixel-value representations and texts are transformed into a form aligned with the corresponding images (Li et al., 2023b). The input image is normalized and divided into patches by the AutoProcessor and then fed into a ViT-based visual encoder. The encoder generates a set of vector representations for each patch as formulated in Equation (11).

$$I \rightarrow X_{img} = [x_1, x_2, x_3, \dots, x_{N_p}], x_i \in \mathbb{R}^{d_v} \quad (11)$$

In Equation (11), I denotes the input image, X_{img} represents its patch-wise visual embedding sequence. Each vector x_i corresponds to the d_v -dimensional feature representation of the i -th image patch. Following this step, the Vision Transformer (ViT) layers apply a multi-head attention mechanism to obtain the visual feature vector V_{img} . These layers are kept frozen and are not updated during LoRA training (Cho et al., 2025).

In the BLIP-2 architecture, the Q-Former transfers representations from the visual encoder to the linguistic space through learnable query vectors. Within each attention block, the standard multi-head attention operation is performed as defined in Equation (12).

$$Attn(Q, V, K) = softmax\left(\frac{(QW_Q)(KW_K)^T}{\sqrt{d_k}}\right)(VW_V) \quad (12)$$

In Equation (12), Q, K , and V denote the query, key, and value matrices, respectively. W_Q, W_K , and W_V are the corresponding learned projection weights. The parameter d_k is the dimensionality of the key vector used for scaling in the attention computation. In this study, to improve parameter efficiency, the attention weight matrices were extended with LoRA components (Hu et al., 2021). LoRA adapters are applied exclusively to the projection matrices of the Q-Former. The visual encoder and the language decoder remain entirely frozen, and no LoRA modules are inserted into their layers. This design follows the parameter efficient tuning strategy of BLIP-2 and ensures that only the Q-Former is updated during training, while the remaining components preserve their

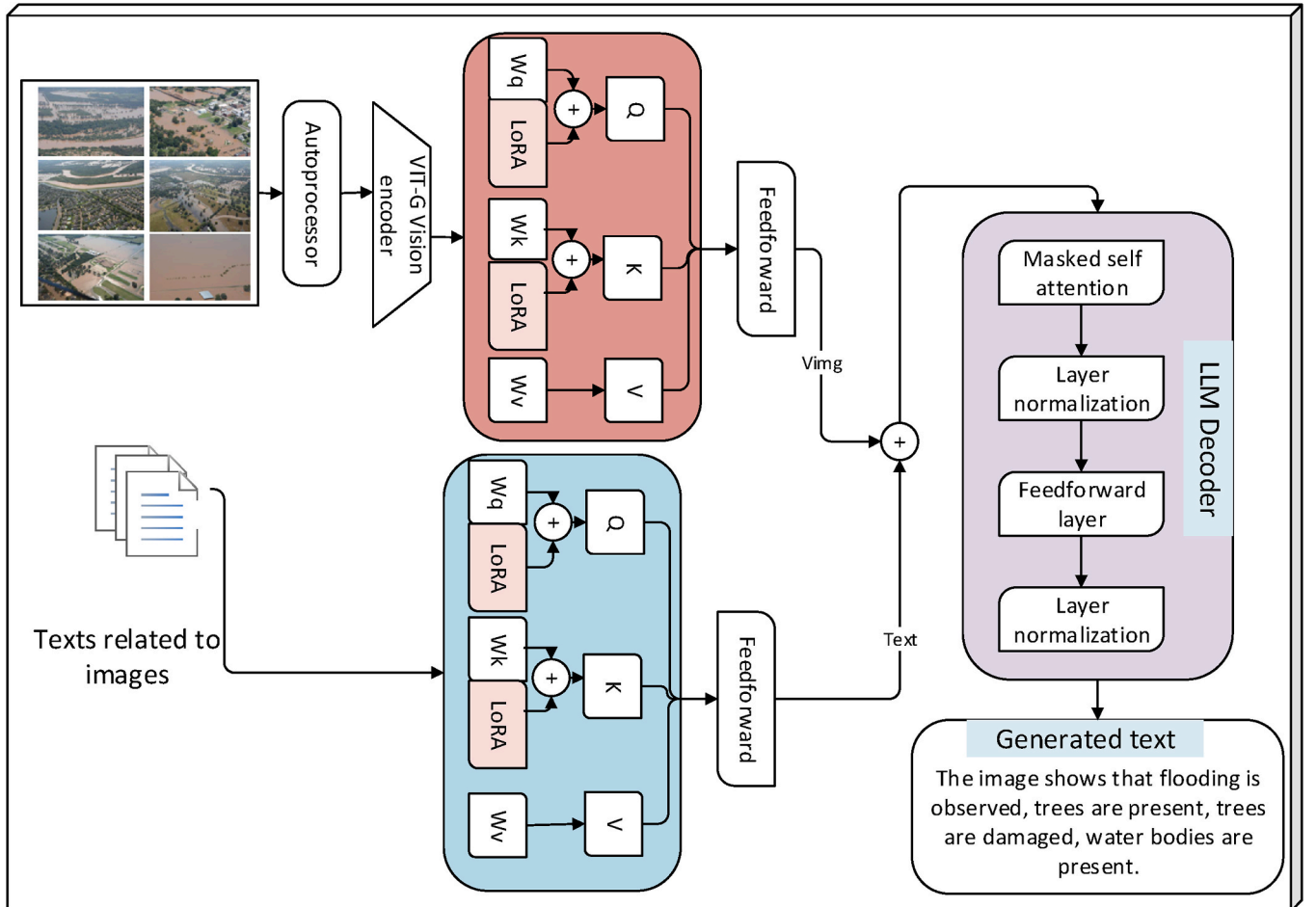


Fig. 4. BLIP-2 and LoRA-Based flood image captioning model.

pre-trained representations. A low-rank update was added to each projection matrix to enable efficient fine-tuning with fewer trainable parameters. The applied modifications are presented in Equation (13).

$$W_Q = W_Q^{(0)} + A_Q B_Q, W_K = W_K^{(0)} + A_K B_K, W_V = W_V^{(0)} + A_V B_V \quad (13)$$

In Equation (13), the matrices $A \in \mathbb{R}^{d_{\text{tr}} \times r}$ and $B \in \mathbb{R}^{r \times d_{\text{tr}}}$ are low-rank LoRA parameters applied to each projection matrix. Here, the rank parameter r is chosen such that $r \ll d$, ensuring that only a small number of additional parameters are introduced. This low-rank structure enables efficient fine-tuning while keeping the original pre-trained weights $W_Q^{(0)}$, $W_K^{(0)}$, and $W_V^{(0)}$ frozen. The Q-Former output produces the component W_{img} through a feed-forward layer that fuses visual features with linguistic queries. The resulting visual-language representations are then passed to a pre-trained language decoder model, which, at each decoding step, computes the probability of the next word according to Equation (14).

$$P(y_t | y_{<t}, I) = \text{softmax}(W_0 h_t) \quad (14)$$

In Equation (14), y_t represents the token being generated, $y_{<t}$ denotes the sequence of preceding words, and I refers to the input image. Additionally, W_0 is the weight matrix that projects the decoder output into the vocabulary space, while h_t corresponds to the hidden state of the decoder at time step t . The model's objective loss function is based on the standard cross-entropy formulation, which is defined in Equation (15).

$$\mathcal{L}_{CE} = - \sum_{i=1}^L \log P(y_t^{\text{true}} | y_{<t}, I) \quad (15)$$

In Equation (15), L denotes the sequence length, y_t^{true} is the ground-truth token at time step t , and $\mathcal{L}_{CE}$ represents the cross-entropy loss computed over the full sequence. A regularization term is added to the loss function to constrain the norms of the LoRA matrices, as formulated in Equation (16). This regularization helps prevent overfitting by controlling the magnitude of the learnable low-rank parameters.

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \sum_i \left(\|A_i\|_F^2 + \|B_i\|_F^2 \right) \quad (16)$$

In Equation (16), λ is the regularization coefficient, while $\|A_i\|_F^2$ and $\|B_i\|_F^2$ denote the Frobenius norms of the LoRA low-rank matrices A_i and B_i , respectively. The total loss $\mathcal{L}$ combines the cross-entropy objective with a norm-based penalty to control the magnitude of the learnable low-rank parameters.

In the final stage of the model, the structural scene information obtained from the segmentation output is combined with the textual descriptions generated in this stage to produce a comprehensive disaster report. The segmentation module generates numerical or proportional statistics for each flood-related class present in the image. These structural outputs are then semantically integrated with the text-based descriptions produced by the decoder of the BLIP-2 model, resulting in a unified and interpretable representation of the disaster scene.

2.3. Evaluation metrics

In this study, two distinct groups of metrics were employed to quantitatively evaluate the performance of both the image segmentation model and the vision-language model. For the segmentation models, three primary metrics were used: Mean Intersection over Union (mIoU), Mean Pixel Accuracy (mPA), and the F1-score. These metrics are widely adopted in segmentation tasks as they provide a detailed assessment of the model's performance in terms of both class discrimination and spatial accuracy.

Each metric is mathematically defined below. The Intersection over

Union (IoU) measures the overlap between the predicted region and the ground truth. It is calculated individually for each class, and its mean value across all classes yields the mIoU, as expressed in Equation (17).

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_j a p_{ij} + \sum_j a p_{ji} - p_{ii}} \quad (17)$$

In Equation (17), k denotes the number of classes in the segmentation task. For each class i , p_{ii} represents the number of correctly predicted pixels. The first summation term corresponds to all pixels predicted as class i , while the second summation represents all ground-truth pixels belonging to class i . The denominator therefore forms the union for class i by combining all predicted and actual pixels and subtracting the correctly predicted pixels to avoid double counting. The final mIoU value is obtained by averaging the IoU of each class across all $k+1$ classes. Pixel Accuracy (PA) is defined as the ratio of correctly classified pixels for a given class to the total number of pixels belonging to that class. The Mean Pixel Accuracy (mPA) is obtained by averaging this value across all classes. It is calculated as shown in Equation (18).

$$mPA = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij}} \quad (18)$$

In Equation (18), p_{ij} denotes the number of pixels whose ground-truth class is i but that are predicted as class j . The F1-score is defined as the harmonic mean of recall and precision. It is a robust metric that reflects the overall balance by simultaneously considering false positives and false negatives, especially in tasks with class imbalance. It is calculated as shown in Equation (19).

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (19)$$

On the vision-language model side, after fine-tuning with the BLIP-2 LoRA method, the quality of the generated descriptions was evaluated using the BLEU, METEOR, and CIDEr metrics. These metrics measure the model's linguistic coherence and its closeness to reference captions, reflecting both syntactic and semantic accuracy. The BLEU metric quantifies the similarity between the model-generated caption and the reference caption (Papineni et al., 2002). The BLEU score is computed as shown in Equation (20).

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (20)$$

In Equation (20), p_n denotes the n-gram precision, while w_n and BP represent the weight coefficient and the brevity penalty term, respectively. The METEOR metric, on the other hand, computes a score based on precision and recall, taking into account word alignment, stemming, and synonym matching to provide a more semantically grounded evaluation (Banerjee and Lavie, 2005). The calculation of the METEOR score is presented in Equation (21).

$$METEOR = \frac{10 \cdot P \cdot R}{R + 9P} \left(1 - \frac{1}{2} \left(\frac{\# \text{ of chunk}}{\# \text{ of matching unigram}} \right)^3 \right) \quad (21)$$

In Equation (21), “# of chunks” represents the number of matched word groups between the reference and predicted sentences, while “# of matching unigrams” indicates the total number of correctly aligned words. The values P and R correspond to precision and recall, respectively.

The CIDEr metric evaluates the similarity between human-generated and model-generated captions using a TF-IDF weighted n-gram similarity approach (Vedantam et al., 2015). This measure is formally defined in Equation (22).

$$CIDEr_n(c_t, S_t) = \frac{1}{|S_t|} \sum_{s \in S_t} Sim_n(c_t, s) \quad (22)$$

Here, c_t denotes the generated text, and S_t represents the reference text, while Sim_n indicates the cosine similarity between the TF-IDF vectors of the n-grams.

3. Experimental results

For the performance evaluation of the proposed two-stage approach, two datasets were utilized. The FloodNet dataset consists of high-resolution UAV images collected to analyze structures and changes in post-flood disaster scenes (Rahneemoonfar et al., 2021). It provides a diverse set of challenging scenarios with annotated frames for both segmentation and classification tasks. In this study, FloodNet was used to assess the model's capability in spatial detection and class differentiation.

The dataset contains a total of 2343 images, labeled for 10 semantic classes: background, flooded building, non-flooded building, flooded road, non-flooded road, water, tree, vehicle, pool, and grass. This dataset was employed to evaluate the performance of the MASA-based image segmentation approach. The FloodNet dataset was divided into training, validation, and test subsets using a random splitting strategy implemented through the PyTorch DataLoader pipeline. Specifically, 70 % of the data was allocated for training, 15 % for validation, and 15 % for testing. The split was performed at the image level using a fixed random seed to ensure reproducibility across experiments. All models were trained and evaluated using the same data partitions, guaranteeing fair comparison and consistent generalization assessment.

The second dataset used in this study is LADI-v2 (Low-Altitude Disaster Imagery), which contains high-resolution aerial images captured from low altitudes following flood events (Scheele et al., 2025). It has a multi-label structure, consisting of 10,000 images paired with descriptive textual annotations. The labeling scheme in LADI-v2 focuses on identifying the presence of specific objects or events within each scene. Accordingly, each image is annotated across multiple categories such as bridge, building, severely flooded building, partially affected building, debris, flood area, flood structures, road, damaged road, tree, flood-affected tree, and water surface.

For linguistic label generation, if an object of a particular class is present in the image, a corresponding descriptive sentence is generated. For example, if a building object exists, the sentence "The buildings are present" is added. Using these two datasets, the system produces both a textual disaster report and detailed information on affected roads, bridges, buildings, and trees.

To evaluate the effectiveness of the proposed MASA segmentation

model, extensive experiments were conducted on the FloodNet dataset (Rahneemoonfar et al., 2021). The performance of MASA was quantitatively compared with a broad set of representative semantic segmentation architectures. These include CNN-based models such as U-Net (Ronneberger et al., 2015), U-Net++ (Zhou et al., 2018), LinkNet (Chaurasia and Culurciello, 2017), PAN (Li et al., 2018), FPN (Lin et al., 2017), DeepLabV3+ (Chen et al., 2018), and UPerNet (Xiao et al., 2018), evaluated with different backbone configurations (ResNet-34 and ResNet-50). In addition, transformer-based approaches were considered, including SegFormer (Xie et al., 2021) with MiT-B1 and MiT-B2 backbones, MaskFormer (Cheng et al., 2021) with Swin-T and ResNet-50 backbones, and SETR (Zheng et al., 2021) using a ViT-L encoder. Model performance was assessed using standard semantic segmentation metrics, namely mean Pixel Accuracy (mPA), F1-score, and mean Intersection over Union (mIoU), enabling a comprehensive and fair comparison between convolutional and transformer-based architectures under identical experimental settings.

All models were trained with an input size of 512×512 and a batch size of 2, using the Adam optimizer with a learning rate of 0.0001 and the sparse categorical cross-entropy loss function. Training was performed for a maximum of 100 epochs, while early stopping was applied based on validation performance to prevent overfitting. As a result, the training process was automatically terminated once convergence was reached, and not all models necessarily completed the full 100 epochs. All experiments were conducted on a system equipped with an Intel Core i9 processor, 128 GB RAM, and an NVIDIA RTX A5000 GPU (24 GB), running Windows 11. Table 2 presents the experimental results obtained from the evaluated segmentation models within the scope of this study.

Table 2 gives a detailed class-wise comparison of the proposed MASA model with the best CNN-based and transformer-based segmentation architectures on the FloodNet dataset. The results show that MASA achieves the highest overall performance, with an mIoU of 73.78 %, outperforming all competing methods under the same experimental conditions. Notably, MASA shows a clear advantage in flood-critical classes: the IoU reaches 88.55 % for the building-flooded class and 80.31 % for the road-flooded class, both of which are essential for post-disaster assessment and emergency response. In addition, MASA demonstrates strong performance in the vehicle and pool categories, achieving IoU scores of 75.67 % and 83.21 %, respectively, which indicates its capability to identify small and partially occluded objects in complex flood scenarios. Although several transformer-based models achieve competitive performance in background and vegetation-related classes, MASA consistently delivers superior results across flood-affected structural categories. This performance highlights the effectiveness of the proposed multi-scale attention-driven feature aggregation strategy. Overall, these findings confirm that MASA effectively balances global

Table 2
Comparison of models IoU performance.

Model	Backbone	BG	Building-flooded	Building-non-flooded	Road-flooded	Road-non-flooded	Water	Tree	Vehicle	Pool	Grass	MIoU (%)
DeepLabV3+	Resnet34	58.41	60.95	78.94	52.39	82.62	74.27	81.59	50.96	60.41	88.13	68.87
DeepLabV3+	Resnet50	52.19	58.52	77.58	45.67	82.15	70.37	79.26	38.95	52.68	87.19	64.46
UNet	Resnet34	40.88	60.03	77.1	50.69	81.03	71.28	81.42	0.0	0.0	87.45	54.99
Unet++	Resnet34	56.80	62.32	79.87	49.95	82.94	71.62	81.44	51.73	62.82	87.6	68.71
Linknet	Resnet34	81.04	71.69	63.89	58.26	51.08	52.97	48.68	53.74	76.11	69.78	62.72
FPN	Resnet34	52.63	60.93	78.73	49.81	82.26	74.04	81.41	48.73	59.55	88.11	67.62
SegFormer	Mit-B1	40.26	59.70	78.46	50.03	80.76	70.24	81.02	40.28	54.12	87.65	64.25
SegFormer	Mit-B2	52.16	59.50	75.98	50.85	82.27	71.54	82.14	43.77	55.55	87.98	66.17
UPerNet	Resnet34	43.06	59.58	75.61	51.23	81.13	71.23	81.01	41.40	54.50	87.97	64.67
UPerNet	Resnet50	52.74	54.55	78.64	46.52	81.04	71.15	80.24	42.13	56.67	87.34	65.10
PAN	Resnet34	54.11	61.13	78.0	49.84	82.53	73.06	81.09	46.76	58.2	88.0	67.33
PAN	Resnet50	47.76	56.68	73.46	48.69	79.49	70.85	79.85	34.71	49.8	87.09	62.84
MaskFormer	SwinT	46.20	78.13	80.7	62.43	82.87	72.64	84.51	22.28	22.88	89.60	64.23
MaskFormer	Resnet50	29.20	77.97	78.43	52.80	82.51	67.68	83.53	22.62	21.85	88.48	60.51
SETR	Vit-L	43.10	68.68	73.13	60.11	78.39	69.38	78.68	21.16	45.29	86.05	62.40
MASA	Resnet50	70.45	88.55	73.32	80.31	62.38	68.69	61.34	75.67	83.21	73.86	73.78

Table 3
Comparison of models based on evaluation metrics.

Model	Backbone	mPA(%)	F1(%)	MIoU(%)
DeepLabV3+	Resnet34	79.74	76.30	68.87
DeepLabV3+	Resnet50	75.51	74.25	64.46
Unet	Resnet34	64.31	62.55	54.99
Unet++	Resnet34	80.16	68.26	68.71
Linknet	Resnet34	76.45	66.20	62.72
FPN	Resnet34	79.01	76.03	67.62
SegFormer	Mit B1	75.17	73.12	64.25
SegFormer	Mit B2	77.39	76.06	66.17
UPerNet	Resnet34	75.73	74.26	64.67
UPerNet	Resnet50	76.86	74.36	65.10
PAN	Resnet34	64.31	73.37	67.33
PAN	Resnet50	73.94	70.86	62.84
MaskFormer	SwinT	73.89	69.06	64.23
MaskFormer	Resnet50	70.01	68.43	60.51
SETR	ViT-L	72.35	74.89	62.40
MASA	Resnet50	81.57	77.23	73.78

contextual modeling with detailed spatial representation, making it particularly well suited for semantic segmentation in flood scenes.

The proposed MASA model was evaluated for practical use through its architectural efficiency assessment in addition to its segmentation accuracy evaluation. Since the number of learnable parameters and backbone complexity directly affect computational cost and deployment feasibility, MASA was compared with a diverse set of widely adopted CNN-based and transformer-based architectures, including U-Net, U-Net++, LinkNet, PAN, FPN, DeepLabV3+, UPerNet, SegFormer, MaskFormer, and SETR. The results in Table 3 demonstrate that MASA achieves a favorable trade-off between segmentation performance and architectural efficiency, delivering superior accuracy while avoiding the excessive model complexity typically associated with transformer-based designs. The system achieves an equilibrium between performance and processing speed. This makes it ideal for actual flood monitoring operations and emergency disaster response activities.

Moreover, evaluating model performance through the visual quality of the generated segmentation outputs is of critical importance for assessing real-world applicability. Table 4 presents a qualitative comparison of the segmentation results produced by the proposed MASA model on the FloodNet dataset against representative state-of-the-art semantic segmentation approaches, together with the corresponding Ground Truth (GT) annotations.

The visual results indicate that MASA generates more coherent object boundaries and more accurate delineation of flood-affected regions when compared to both the GT masks and competing methods, particularly for structurally critical classes such as flooded buildings and roads. These qualitative observations are consistent with the quantitative findings and further demonstrate the robustness and practical effectiveness of the proposed MASA architecture. Table 4 shows a qualitative comparison of segmentation outputs on the FloodNet dataset, including the original images, ground-truth annotations, and predictions obtained from different models. In this evaluation, SegFormer uses a MiT-B2 backbone, MaskFormer is implemented with Swin-T, and SETR relies on a ViT-L encoder, while the remaining approaches are based on a ResNet-50 backbone. The visual results suggest that transformer-based models generally capture broader contextual information and generate relatively sharp boundaries; however, they may still produce irregular predictions in regions affected by flooding. Models relying solely on convolutional backbones tend to experience greater difficulty when separating visually similar and overlapping classes, such as buildings and roads in dense urban scenes. Compared to these approaches, the proposed MASA model provides more stable and visually consistent segmentation outputs, particularly in distinguishing flooded and non-flooded structures and road surfaces. This indicates that MASA is better suited for handling the complex spatial patterns commonly observed in flood scenarios.

In the second part of the study, the BLIP-2 model was adopted as the foundation for learning vision linguistic representations, with the LoRA approach integrated to enhance parameter efficiency. The large-scale weights in the language encoder were frozen, while only the projection layers were made trainable. Experiments were conducted on the LADI-v2 dataset, where all images were resized to 512×512 pixels, and the maximum text length was limited to 64 tokens. During training, the AdamW optimizer was employed with a learning rate of 1×10^{-4} , batch size of 2, and gradient accumulation set to 4. All models were trained using bf16 precision for 40 epochs. During inference, the num_beams and length penalty parameters were set to 5 and 1.05, respectively. The model's performance was evaluated using BLEU-4, METEOR, and CIDEr metrics. Fig. 5 illustrates the loss variation observed during the training process of the BLIP-2 + LoRA model.

As shown in Fig. 5, the BLIP-2 + LoRA model exhibits rapid convergence from the early epochs. The performance of the proposed method was evaluated on the LADI-v2 dataset and compared with the well-known BLIP and BLIP-Instruct models from the literature. The

Table 4
Visual comparisons of the proposed model with other models.

Image	GT	SegFormer	PAN	DeepLabV3+	MaskFormer	SETR	MASA
							
							
							
							
							
							



Fig. 5. Loss curve of the BLIP-2 + LoRA fine-tuning process.

Table 5

Comparison of performance among BLIP, InstructBLIP, and BLIP-2+LoRA models.

Model	Number of Parameters (Billion)	Percentage of trainable parameters (%)	BLEU-4	METEOR	CIDEr
BLIP-Base	0.24	100.00	77.90	81.23	6.74
Instruct BLIP + LoRA	4.24	5.10	79.31	84.38	6.51
BLIP-2+LoRA	2.70	5.62	80.77	84.53	8.08

models were compared using the BLEU-4, METEOR, and CIDEr metrics. The comparison results are presented in Table 5.

As shown in Table 5, the BLIP-2 + LoRA model demonstrates the most balanced performance in terms of both linguistic accuracy and parameter efficiency. Compared to the original BLIP model, the InstructBLIP + LoRA and BLIP-2 + LoRA architectures contain fewer trainable parameters. The BLIP-2 + LoRA model achieved superior results for both the BLEU and METEOR metrics, indicating that LoRA-based fine-tuning enables effective learning in large models through lightweight adaptation layers. Overall, the BLIP-2 + LoRA model exhibits high parameter efficiency, strong semantic accuracy, and enhanced vision-linguistic coherence. Fig. 6 presents examples of generated captions and reference texts for images containing different objects.

In Fig. 6, the reference descriptions, shown in blue, represent the ground-truth annotations provided in the dataset, while the predicted descriptions, shown in red, correspond to the linguistic outputs generated by the model. The predicted captions accurately identify key objects such as flood-affected buildings, roads, and trees. Moreover, the model demonstrates the ability to capture semantic details such as the degree of structural damage and the extent of water inundation, indicating its potential to enhance situational awareness in post-disaster scenarios. By combining the results from the vision language model and the MASA segmentation model, a comprehensive reporting framework is established. The segmentation model distinguishes buildings, roads, and natural structures at the pixel level within flood-affected regions, enabling the counting of flooded buildings. Meanwhile, the vision language model generates automatic textual descriptions for the same scenes. This integrated framework goes beyond simple image classification or pixel-level prediction by jointly modeling both physical

and semantic changes in the disaster environment, thus contributing to post-disaster decision support processes. Fig. 7 presents representative examples illustrating the proposed interpretable flood damage assessment pipeline. For each scene, the original RGB image, ground-truth (GT) segmentation mask, and predicted (Pred) segmentation mask are shown together with their corresponding overlay visualizations. Flooded buildings and flooded roads are highlighted in the overlays, and instance-level flooded building counts are obtained.

In Fig. 7, the comparison between GT and predicted overlays explicitly reveals both accurate detections and discrepancies, allowing potential sources of misinterpretation to be visually inspected. In particular, differences in flooded-building counts and the presence or absence of flooded roads between GT and model predictions demonstrate the limitations of automated assessments and the importance of transparency in decision-support systems. In addition, the rightmost panels show natural-language descriptions generated by the BLIP-2+LoRA vision language model, augmented with automatically extracted quantitative findings. While these examples demonstrate the potential of integrating segmentation outputs with language based explanations for post disaster reporting, they are intended as qualitative illustrations rather than definitive validation. Together, these visualizations support the interpretability of the proposed framework by enabling direct comparison between ground truth, model predictions, and generated explanations, thereby facilitating human-in-the-loop analysis in operational flood assessment scenarios.

While the qualitative case studies illustrate the general behavior of the proposed framework combining segmentation outputs, instance separation, and natural-language reporting this visualization alone is insufficient to assess the reliability of instance-level damage estimation. In particular, for decision-support scenarios, it is essential to quantify how accurately the system estimates the number of flooded buildings compared to the ground truth. To address this limitation, we conduct an instance-level flooded-building counting evaluation by directly comparing counts derived from ground-truth masks and model predictions. We compare the number of flooded buildings derived from ground-truth and predicted masks as shown in Fig. 8.

The evaluation is conducted by comparing flooded-building counts derived from ground-truth masks and model predictions for each validation image. Quantitatively evaluate the accuracy of instance-level flooded-building estimation, we employ **mean absolute error (MAE)** and **root mean squared error (RMSE)** as complementary regression-based metrics. MAE provides an intuitive measure of the average deviation between predicted and ground-truth building counts, directly

Ref: The image shows that bridges are present, buildings are present, flooding is observed, roads are present, trees are present, water bodies are present.



Pred: The image shows that bridges are present, buildings are present, flooding is observed, roads are present, trees are present, water bodies are present.

Ref: The image shows that buildings are present, flooding is observed, roads are present, trees are present, water bodies are present.



Pred: The image shows that buildings are present, flooding is observed, roads are present, trees are present, water bodies are present.

Ref: The image shows that flooding is observed, roads are present, trees are present, trees are damaged, water bodies are present.



Pred: The image shows that flooding is observed, roads are present, trees are present, trees are damaged, water bodies are present.

Ref: The image shows that flooding is observed, roads are present, trees are present, trees are damaged, water bodies are present.



Pred: The image shows that flooding is observed, roads are present, trees are present, trees are damaged, water bodies are present.

Fig. 6. Linguistic generation results of the BLIP-2 + LoRA model on sample images from the FloodNet and LADI datasets.

reflecting the expected counting error per scene. In contrast, RMSE penalizes larger errors more heavily, enabling the identification of occasional but potentially critical misestimating that may impact emergency response planning. Together, these metrics offer a balanced assessment of both typical and worst-case counting performance. Table 6 summarizes the instance-level flooded-building counting performance on the validation set using multiple complementary metrics.

Quantitative results indicate that the proposed method achieves a **MAE of 0.38 buildings** and a **RMSE of 1.64 buildings**, demonstrating that the predicted number of flooded buildings deviates by less than one building on average. The observed **bias of 0.195** suggests a very mild tendency toward overestimation, without indicating any systematic counting error.

4. Ablation study

To better understand the contribution of the main building blocks of the proposed model, an ablation study was conducted by progressively integrating the ASPP and Self-Attention modules into the baseline architecture. The quantitative results are summarized in Table 7. All experiments were performed on a system equipped with an NVIDIA RTX 3060 GPU, and the reported inference times correspond to the average forward-pass latency measured under identical experimental conditions. The baseline FCN model achieves an mIoU of 63.90 % with an inference

time of 13.28 ms, serving as a reference for subsequent comparisons. Incorporating the ASPP module improves segmentation performance to 67.52 % mIoU while slightly increasing the inference time to 15.70 ms due to the additional multi-scale context processing. When the Self-Attention module is further introduced, the mIoU significantly rises to 73.78 %, with a marginal increase in inference time to 15.88 ms, indicating that the proposed enhancements achieve substantial accuracy gains with only limited computational overhead.

A more substantial improvement is observed when the Self-Attention mechanism is combined with ASPP, resulting in an mIoU of 73.78 %. This performance gain is achieved with only a slight increase in the number of parameters, computational cost, and inference time. The results suggest that Self-Attention helps the model better capture long-range relationships and maintain semantic consistency across flood-affected regions. Overall, the ablation study shows that using ASPP and Self-Attention together leads to a clear improvement in segmentation accuracy while maintaining reasonable computational efficiency.

In addition, the effect of different dilation rate settings within the ASPP module was further examined, since dilation rates directly influence the receptive field and the quality of contextual feature extraction. Several dilation rate configurations were therefore evaluated to determine the most effective setup, and the corresponding results are presented in Table 8.

Fig. 9 shows the training curves for Loss and Accuracy under

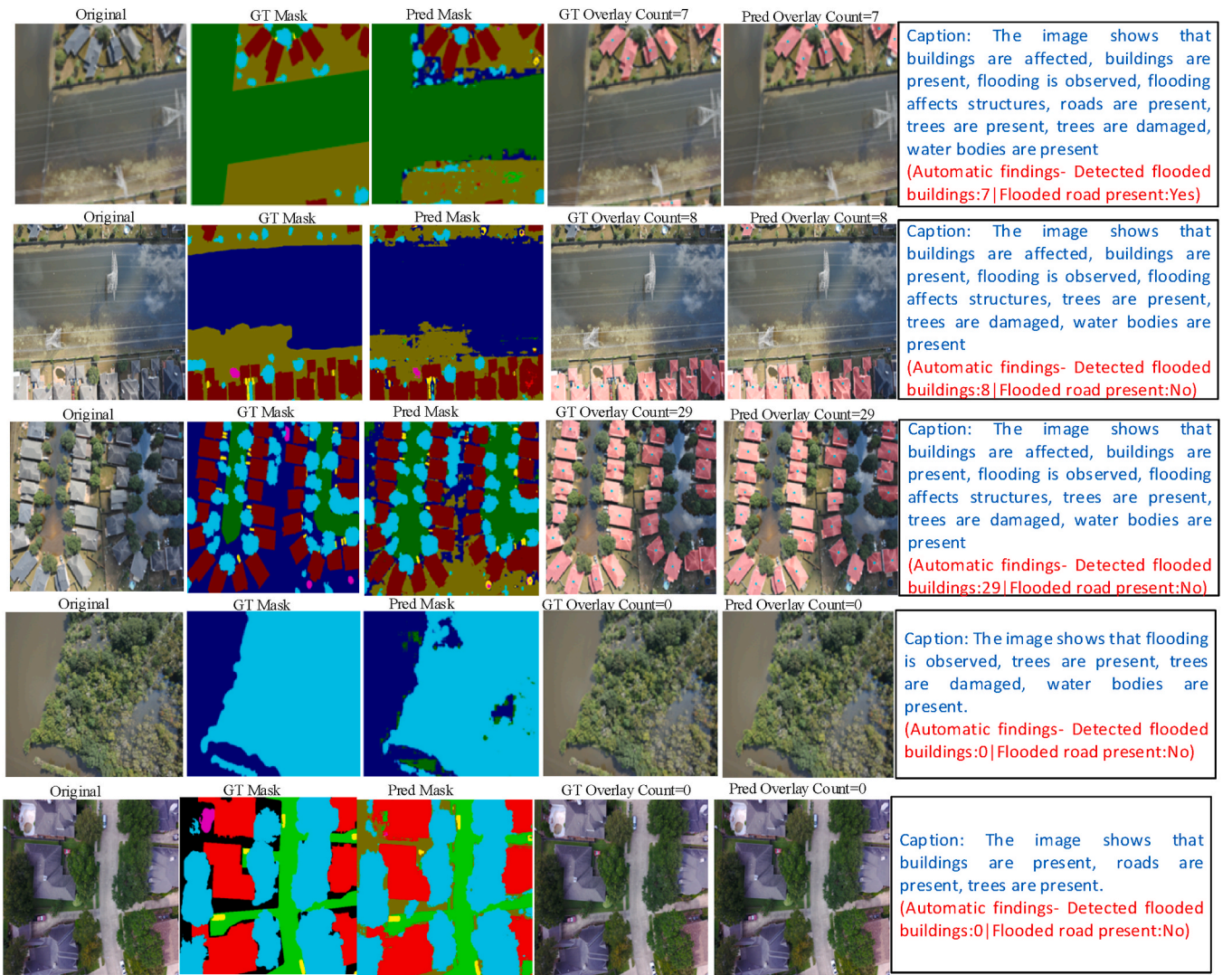


Fig. 7. Segmentation and description integration on a sample scene from the FloodNet dataset.

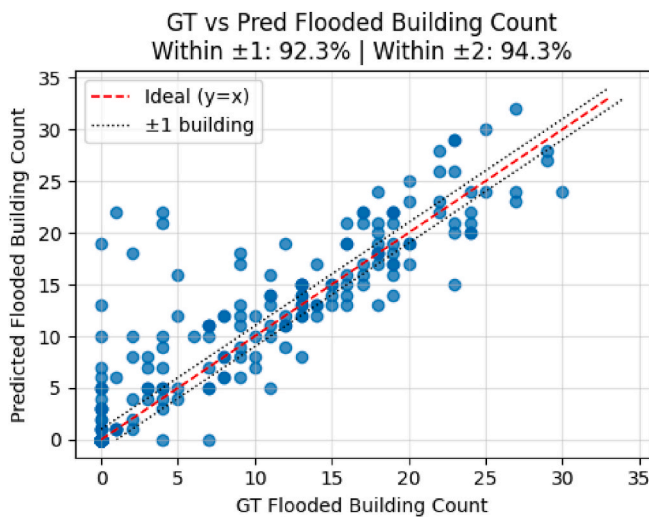


Fig. 8. Comparison of ground-truth and predicted flooded-building counts.

different dilation-rate settings. The [1, 6, 12, 18] setup produced a rapid and stable loss reduction and achieved the highest accuracy, while [1, 3,

Table 6

Quantitative evaluation of instance-level flooded-building counting accuracy.

Metric	Value
MAE	0.379
RMSE	1.639
Bias	0.195
Counting Accuracy (%)	95.79
Within ± 1 Building Accuracy (%)	92.32
Within ± 2 Building Accuracy (%)	94.26

Table 7

Ablation study with different modules.

Configuration	MIoU (%)	Params (M)	FLOPs (G)	Time (ms)
Baseline(FCN)	63.90	10.11	39.03	13.28
Baseline + ASPP	67.52	17.78	46.62	15.70
ASPP + SelfAttention	73.78	17.86	46.71	15.88

6, 9] also reached high accuracy in fewer epochs. In terms of IoU, the [1, 3, 6, 9] configuration delivered the best performance, achieving the highest Final Training IoU of 89.24 %, confirming its suitability for flood segmentation tasks.

Table 8
Different dilation rates used in the ablation study.

Dilation Rate	Feature of Dilation Rate
[1, 6, 12, 18]	Widely used and broadly spaced dilation rates.
[1, 3, 6, 9]	Moderately increasing, balanced structure.
[6, 12, 18, 24]	A structure focusing on large intervals and high dilation rates.
[1, 4, 8, 16]	Moderately scaled dilation rates
[2, 4, 8, 16]	A balanced structure increasing from small to large.

The ablation study on dilation rate settings, summarized in Table 9, provides insight into how different configurations affect segmentation performance. Among the evaluated options, the [1, 3, 6, 9] dilation rate combination yields the most favorable results, achieving an mIoU of 73.78 %, along with a Pixel Accuracy of 81.57 % and an F1 score of 77.23 %. The more widely spaced [1, 6, 12, 18] configuration follows closely with an mIoU of 71.57 %, indicating that larger receptive fields can still be effective in capturing structural context.

In comparison, the [6, 12, 18, 24] setting shows a noticeable drop in performance, reaching an mIoU of 67.53 %, while the more densely spaced [2, 4, 8, 16] rates lead to the lowest result with an mIoU of 65.04 %. Overall, the results suggest that a moderate and balanced increase in dilation rates offers a better trade-off between local detail and contextual information. Based on these observations, the [1, 3, 6, 9] configuration was adopted in the final MASA model.

The selected parameters of LoRA applied to the BLIP-2 model play a crucial role in both learning dynamics and parameter efficiency. In particular, the effects of the rank (r), scaling factor, and dropout rate were analyzed within the developed module to assess their influence on model performance.

Table 10 presents a performance comparison of six experiments in which the LoRA layers were configured with different rank (r), scaling (α), and dropout values. The results include the CIDEr, BLEU, and METEOR metrics, along with the percentage of trainable parameters in the model.

Overall, increasing the r and α values led to a gradual improvement in performance metrics, whereas the impact of increasing the dropout rate remained limited. The best performance was achieved with $r = 32$, $\alpha = 64$, and dropout = 0.05. These findings indicate that higher rank and scaling parameters contribute to more comprehensive information representation in the LoRA configuration, while excessive dropout does not provide additional benefit.

5. Discussion

In this paper, we introduce a multimodal framework designed for

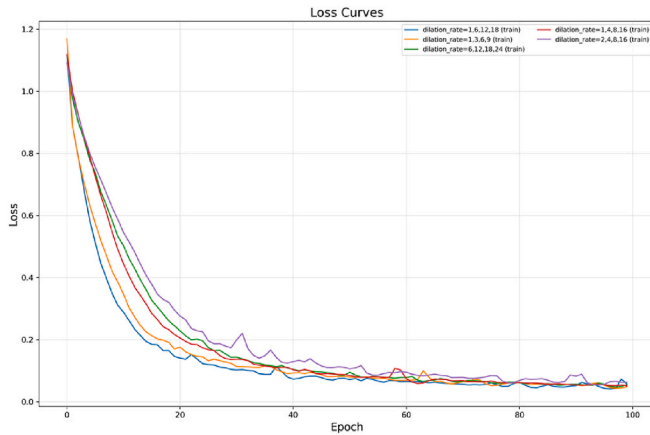
Table 9
Comparative results of different dilation rates in the ASPP module on Mean IoU, Pixel Accuracy, and F1 Score.

Dilation Rates	PA(%)	F1 (%)	mIoU (%)
[1, 6, 12, 18]	77.05	74.96	71.57
[6, 12, 18, 24]	76.18	70.97	67.53
[1, 4, 8, 16]	75.85	73.10	69.69
[2, 4, 8, 16]	70.91	68.26	65.04
[1, 3, 6, 9]	81.57	77.23	73.78

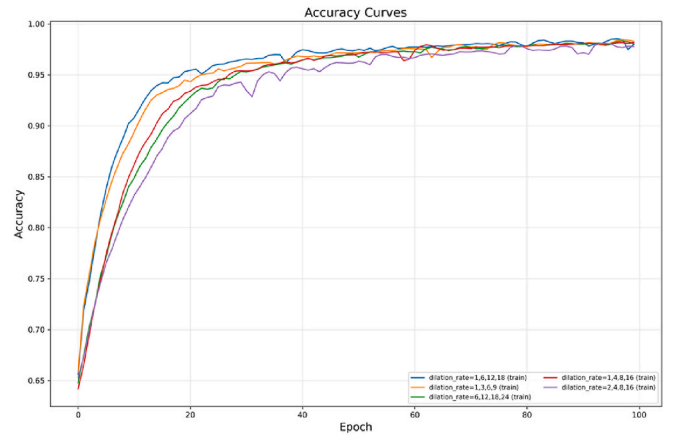
explainable flood damage assessment. By merging a MASA-based semantic segmentation model with a vision language model enhanced via LoRA, we achieved solid results. Specifically, the experimental data shows robust segmentation performance on FloodNet and accurate natural-language descriptions on LADI-v2. The MASA model was evaluated using the FloodNet dataset, which consists of high-resolution UAV imagery captured under post-flood conditions. While the results indicate that the model is highly effective for complex urban flood scenarios, FloodNet represents a specific acquisition setting in terms of sensor type, spatial resolution, illumination conditions, and geographic characteristics. In real emergency situations, image quality may vary significantly due to unfavorable lighting, extreme viewing angles, motion blur, occlusions caused by debris or vegetation, and adverse weather conditions. Such factors may affect the reliability of pixel-level segmentation and should be considered when deploying the proposed system in operational environments.

It is important to clarify that the primary objective of this study was not to develop a universally generalizable segmentation architecture. Instead, we aimed to identify a stable and effective model configuration that can reliably support downstream explainable analysis. Through extensive comparisons with a wide range of state-of-the-art CNN-based and transformer-based models, the selected MASA configuration demonstrated a favorable balance between accuracy, stability, and computational efficiency. This balance is particularly critical for flood-sensitive classes such as flooded buildings and roads, which directly influence damage estimation and emergency decision making.

Regarding generalization across unseen geographical regions, differences in urban layout, building materials, road structures, and environmental context may introduce domain shifts that were not fully captured in the training data. As a result, additional adaptation or validation may be required when applying the framework to flood events occurring in regions with significantly different visual characteristics. For the vision language component, training relied on the LADI-v2 dataset, and the generated descriptions naturally reflect the annotation structure and semantic coverage of that dataset. Although



(a)



(b)

Fig. 9. Effects of different dilation rates in the ASPP module during the training process: (a) Loss graph, (b) Accuracy graph.

Table 10
LoRA ablation experiment results.

Experiment No	r	Alpha	Dropout	Trainable parameters(%)	CIDEr	BLEU	Meteor
1	8	16	0.05	4.98	7.97	78.20	0.83
2	8	16	0.1	4.98	7.95	78.14	0.83
3	16	32	0.05	5.20	8.03	78.82	0.84
4	16	32	0.1	5.20	8.06	79.03	0.83
5	32	64	0.05	5.62	8.08	80.77	0.84
6	32	64	0.1	5.62	8.07	79.72	0.84

the produced textual outputs are coherent and informative, their reliability may be influenced by biases present in the training data, as well as by ambiguous visual cues in complex disaster scenes. In real-world deployments, such descriptions should therefore be interpreted as supportive explanatory information rather than definitive assessments.

Future work will focus on adapting the proposed approach for real-time and field-deployable applications. In particular, both the MASA-based segmentation module and the BLIP-2 LoRA vision-language model will be integrated into UAV-based disaster response systems. Model optimization will also be investigated through lightweight architectures and hardware-aware pruning techniques to enable efficient inference on edge devices. Beyond flood scenarios, the proposed framework will be extended to other natural disasters such as earthquakes, landslides, wildfires, and storms by incorporating disaster-specific segmentation classes and domain-adapted multimodal datasets. On the language modeling side, multilingual and cross-lingual extensions will be explored to support automated disaster reporting in multiple regional languages, thereby improving accessibility and practical usability.

6. Conclusions

This study integrates a MASA based segmentation model with a BLIP-2 LoRA enhanced vision language explanation module for post flood damage assessment. The proposed segmentation model combines the ASPP structure with a self-attention mechanism, effectively capturing both fine details and broad contextual information, achieving highly accurate segmentation in complex flood scenarios. Despite having a relatively small number of parameters, the MASA architecture outperforms many conventional segmentation models.

Experimental evaluations show that the model reliably identifies flood affected structures, particularly buildings and roads, and maintains consistent performance under varying environmental conditions. The ablation study further reveals that moderately increasing the dilation rates in the ASPP module enhances multi scale feature extraction and positively contributes to segmentation performance. Moreover, integrating the self-attention mechanism enables the analysis of spatially distant but semantically related regions, thereby improving overall segmentation quality.

Experimental analyses conducted on the FloodNet dataset confirm the model's high accuracy in detecting flood affected structures. Additionally, the LoRA based fine tuning strategy significantly improved the parameter efficiency of the BLIP-2 model. The BLIP-2 LoRA ablation studies demonstrate that high textual generation performance can be achieved with a limited number of trainable parameters.

As a result, the system not only performs visual level flood region detection but also generates natural language descriptions of the observed situations. The developed automatic reasoning module integrates quantitative outputs such as the number of flooded buildings and the condition of roads into the natural language explanations produced by the BLIP-2 LoRA model.

Overall, the proposed approach exhibits strong performance in image text integrated flood analysis, parameter efficiency, and user interpretability, offering a powerful and explainable framework for post disaster assessment and decision support.

CRedit authorship contribution statement

Ilhan Aydin: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Data curation, Conceptualization. **Emre Güçlü:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Taha Kubilay Şener:** Writing – original draft, Visualization, Software, Methodology. **Erhan Akin:** Writing – review & editing, Validation, Supervision, Conceptualization.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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